SOLUTION DESIGN DOCUMENT

NQR Reconciliation Automation

Blue Prism ? Business Day API ? SQL Reconciliation Engine ? spNQRRecon ? Multi-Category Output

Document Information	
Field	**Details**
Document Reference	SDD-BN-005 - NQR Reconciliation Automation
Version	1.0 - Initial Release
Author	Blessing Nnabugwu, RPA Developer
Date	October 2024
Platform	Blue Prism 7.1.0
Product Code	NQRR
Schedule	Daily - 08:10 AM (business days only)
Status	APPROVED - In Production

Version History			
Version	**Date**	**Author**	**Description**
1.0	October 2024	Blessing Nnabugwu	Initial draft - approved for production

1. Introduction

1.1 Scope

This Solution Design Document describes the architecture, design, and operational details of the NQR Reconciliation Automation - a Blue Prism bot that automates the daily reconciliation of NQR (National QR Code Payment) settlement records against General Ledger (GL) entries.

The bot retrieves NQR settlement report files and GL transit account data, executes a SQL-based reconciliation engine (spNQRRecon) that matches, classifies, and exceptions all transactions, generates a multi-sheet Excel reconciliation workbook across nine output categories, and distributes results to stakeholders.

A key architectural feature of this bot is the database-level reconciliation logic - rather than performing matching in Blue Prism memory, the bot delegates all reconciliation computation to the stored procedure spNQRRecon. This enables high-performance processing of large transaction volumes and ensures reconciliation rules are centralised and auditable.

1.2 Aim

- To describe the functional and technical structure of the automated reconciliation solution.
- To identify required system resources, dependencies, and configuration requirements.
- To provide a reference for assessing the impact of future requirement changes.
- To support the production of UAT test cases and compliance verification.
- To aid in ongoing maintenance, support, and operational handover activities.

1.3 Intended Audience

- RPA Developers and Solution Architects
- Finance and Reconciliation Operations Teams
- Database Administrators
- Audit and Compliance Teams
- IT Infrastructure and Support Teams

1.4 Definitions and Acronyms

Term	Definition
NQR	National QR Code Payment - a payment scheme enabling QR-code-based merchant transactions.
GL	General Ledger - the bank's core accounting system recording all financial entries.
spNQRRecon	The SQL stored procedure that orchestrates the full reconciliation logic - matching, exception classification, auto-reversal detection, and result population.
Transit Account	The GL transit account through which NQR settlement entries are posted - the source of GL data for reconciliation.
Open Item	A transaction that could not be reconciled - present in the GL but not in the settlement report, or vice versa.
Auto Reversed	A GL entry that has been automatically reversed - matched credit and debit entries with the same transaction remarks, excluded from the primary matching run.
NQRR	Product code uniquely identifying the NQR Reconciliation process within the BN Operations Process.

Business Day API	Internal API confirming whether the current date is a valid business day.
BNRPA DB	The SQL Server database used by the bot for data storage, staging, and reconciliation computation.

2. Process Overview

2.1 Business Context

NQR reconciliation is a daily compliance requirement - ensuring that every QR-code payment settled by the scheme is accurately reflected in the bank's General Ledger. Prior to automation, the reconciliation team manually pulled settlement report files, ran comparisons against GL extracts, classified exceptions, and compiled multi-sheet Excel reports. This was a 45-60 minute manual exercise prone to classification errors and inconsistency.

The automation reduces this to approximately 10 minutes end to end - from file retrieval through SQL reconciliation engine execution to stakeholder report delivery. All reconciliation logic is encoded in the stored procedure spNQRRecon, ensuring consistent application of matching rules across every daily run.

2.2 AS-IS Pain Points

Pain Point	Business Impact
Manual GL extract and settlement file comparison	45-60 min daily, high error risk on large volumes
Inconsistent exception classification	Different analysts classify exceptions differently
No automated auto-reversal detection	Reversed entries incorrectly appear as exceptions
Manual multi-sheet report compilation	Time-consuming, formatting errors common
No audit trail of reconciliation runs	Cannot trace which entries were matched on which date
Weekend aggregation handled inconsistently	Non-business day files missed or double-counted

2.3 Reconciliation Schedule

Process	Product Code	Scheduled Start	Avg Duration
NQR Reconciliation	NQRR	08:10 AM daily	~10 minutes

NQR Reconciliation runs on business days only - excluding the business day immediately following a non-working day. Settlement files for non-business days (including the succeeding business day's settlement) are aggregated and reconciled together against the GL dataset obtained for that period. This means the GL dataset will cover transactions initiated across the non-business day period.

The number of NQR settlement report files per run varies: 5 files on a standard business day, 11 files after a weekend (covering Saturday, Sunday, and the preceding business day's carryover sessions).

2.4 Reconciliation Output Categories

The spNQRRecon stored procedure classifies every transaction into one of nine output categories, each written to a dedicated worksheet in the reconciliation workbook:

Category	Description
Proof	Summary proof of the reconciliation - total settlement vs GL balance verification.
Matched	Transactions successfully matched between the NQR settlement report and the GL credit entries.

Exceptions	Settlement report entries that could not be matched to a corresponding GL entry.
GL Balance	GL entries from the transit account - credit and debit totals for balance verification.
Settlement Summary	Summary statistics across all settlement sessions included in the reconciliation run.
Settlement Report	Full settlement report entries as retrieved from the NQRTransactionReport staging table.
Auto Reversed	GL entries identified as auto-reversals - matched credit and debit pairs excluded from primary matching.
GL Exceptions	GL entries (credit and debit) that could not be matched to a settlement report entry.
Duplicates	Duplicate GL entries detected in the NQRRECONGL table for the reconciliation period.

2.5 High-Level Process Flow

1. At the scheduled time, the Process Administrator triggers the BN Operations Process via the Blue Prism scheduler at 06:30 AM.
2. The BN Operations Process calls the Business Day API to confirm whether the current date is a valid business day. If not - the process terminates without attempting reconciliation.
3. The bot queries the BNRPA database to retrieve the NQRR product configuration and confirm eligibility to run.
4. The Get Modified File Name stage computes the dynamic output file path based on the reconciliation date.
5. The Set Working Day stage calculates the working day - adjusting for weekends and holidays - and retrieves the last working business day date.
6. The bot creates the output directory if it does not already exist.
7. The Truncate Report Table stage clears the NQRTransactionReport staging table of previously inserted entries.
8. The Move Files page checks for all required settlement report files in the shared folder. Depending on the business day scenario, 5 or 11 NQR report files are expected. Each file is read and bulk-inserted into the NQRTransactionReport staging table.
9. The Get GL For Transit Account page truncates the NQRRECONGL table and reloads it with GL entries for the transit account for the current reconciliation period.
10. The Validate Spooled GL page validates the completeness and accuracy of the loaded GL entries - checking balance and processing status.
11. If the GL is balanced and valid - the bot executes the stored procedure spNQRRecon, passing the start and end dates. This procedure performs all matching, auto-reversal detection, and exception classification, populating the result tables.
12. The bot executes SQL queries to retrieve results from each of the nine output tables into Blue Prism collections.
13. The Generate Report page writes all collections to a single Excel workbook with one worksheet per output category.
14. The bot sends a stakeholder email with the reconciliation workbook attached.

3. Solution Architecture

3.1 Architectural Overview

The NQR Reconciliation bot uses a database-centric architecture - unlike bots that perform business logic in Blue Prism workflows, this bot delegates all reconciliation computation to the SQL stored procedure spNQRRecon. Blue Prism orchestrates the data preparation and output generation; the database engine performs the matching.

This design provides three key benefits: reconciliation rules are centralised and version-controlled in the database; processing performance scales with the database server rather than the bot machine; and the full reconciliation logic can be audited, tested, and updated independently of the Blue Prism process.

Layer	Component	Role
Orchestration	Process Administrator	Triggered at 06:30 AM - calls BN Operations Process.
Routing	BN Operations Process	Business Day API ? product routing ? NQRR session management.
Processing	NQR Reconciliation VBO	Five-page VBO - data preparation, GL loading, validation, reconciliation, report generation.
Reconciliation Engine	spNQRRecon (SQL Stored Proc)	Full matching logic - auto-reversal detection, GL vs settlement matching, exception classification.
Staging	NQRTransactionReport table	Settlement report entries loaded from files - truncated and reloaded each run.
GL Data	NQRRECONGL table	GL transit account entries - truncated and reloaded each run.
Results	9 result tables	NQRReconMatched ? NQRRECONGLException ? NQRRECONAutoReversed ? plus summary tables.
Output	Excel workbook + Email	9-sheet reconciliation workbook distributed to stakeholders.
External API	Business Day API	Returns IsBusinessDay flag - prevents reconciliation on non-business days.

3.2 BN Operations Process - Main Page Stages

S/N	Stage Name	Type	Description
1	Get Business Day Details	Action Stage	Calls the Business Day API. Returns IsBusinessDay flag for the current reconciliation date.
2	List Details	Multi-Calculation	Assigns API response values to process data items.

3	Is Business Day?	Decision Stage	If False - sends notification email and terminates. If True - continues.
4	Send Email	Action Stage	Sends non-business day notification to stakeholders.
5	Get Products to Run	Page Reference	Queries BNRPA DB for eligible products: not completed, retries not exceeded, scheduled time passed.
6	Loop Through Products	Loop	Iterates through eligible product sessions.
7	Product Routing	Choice Stage	Routes NQRR product code to the NQR Reconciliation Business Object.
8	IsCompleted	Data Item	Stores the completion flag returned by the reconciliation run.
9	Mark Product as Complete	Action Stage	Updates BNRPA database to mark NQRR as successfully completed for the current day.
10	Update Trial Count	Action Stage	Updates BNProductList with the number of processing attempts made.
11	Recover	Recover Stage	Catches exceptions - prevents the entire process from crashing on a single product failure.
12	Resume	Resume Stage	Redirects the process to continue after a handled exception.

4. NQR Reconciliation Business Object

4.1 Overview

The NQR Reconciliation Business Object orchestrates the full reconciliation lifecycle across five functional pages. The main page (Process NQR Recon) coordinates the overall flow - calling sub-pages for data preparation, GL loading, validation, and report generation, then executing the reconciliation stored procedure and retrieving results.

Page	Purpose
Process NQR Recon (Main)	Master orchestration page - date calculations, staging table management, stored procedure execution, result retrieval, and report generation.
Move Files	Checks shared folder for all required settlement report files (5 standard / 11 post-weekend) and bulk-inserts each into the NQRTransactionReport staging table.
Get GL For Transit Account	Truncates the NQRRECONGL table and reloads it with GL transit account entries for the current reconciliation period.
Validate Spooled GL	Validates completeness and accuracy of loaded GL entries - checks balance status and processing success before reconciliation proceeds.
Generate Report	Compiles all nine output collections into a single Excel workbook with one dedicated worksheet per reconciliation category.

4.2 Main Page Input Parameters

Input Parameter	Data Type	Description
ReconDate	Date	Date of the reconciliation instance - passed from Process Administrator via Now().
ProductCode	Text	Database reference code - NQRR.

4.3 Main Page Stages

S/N	Stage Name	Type	Description
1	Get Modified File Name	Action Stage	Computes the dynamic output file path by modifying OutputFilePath based on the ReconDate.
2	Set Working Day	Multi-Calculation	Calculates the working day based on ReconDate - adjusts for weekends and holidays using the last valid working date.
3	Get Last Working Business Day Date	Action Stage	Retrieves the date of the last working business day - used to determine the correct date range for GL and settlement file retrieval.

4	Create Output Directory	Action Stage	Creates the output directory at OutputFilePath if it does not already exist.
5	Truncate Report Table	Action Stage	Executes a TRUNCATE against the NQRTransactionReport staging table - clears all previously inserted entries before the current run's files are loaded.
6	Move Files	Page Reference	Calls Move Files page - checks shared folder for all required NQR settlement report files and bulk-inserts each into NQRTransactionReport.
7	Get GL For Transit Account	Page Reference	Calls Get GL page - truncates NQRRECONGL and reloads with GL entries for the transit account for the current period.
8	Validate Spooled GL	Page Reference	Validates GL entries - checks completeness, balance, and that closing balance is available.
9	Is Balanced and Success?	Decision Stage	Checks if GL is balanced and processed successfully with closing balance available. If not - exception email sent.
10	Execute spNQRRecon	Action Stage	Executes the stored procedure spNQRRecon(@start_date, @end_date) - performs full reconciliation: auto-reversal detection, GL-vs-settlement matching, exception classification. Results are saved in a collection: Result_Exception.
11	Get Matched Items	Action Stage	Executes SQL query against NQRReconMatched - retrieves successfully matched transactions. Saves to collection: Result_Matched.
12	Get Auto Reversed	Action Stage	Executes SQL query against NQRRECONAutoReversed - retrieves auto-reversed GL entry pairs. Saves to collection: AutoReversed.
13	Get GL Exceptions	Action Stage	Executes SQL query against NQRRECONGLException - retrieves unmatched GL credit and debit entries. Saves to collection: GL_Exceptions.
14	Get NQR Report	Action Stage	Executes SQL query against NQRTransactionReport - retrieves full settlement report entries. Saves to collection: NQR_Settlement_Report.
15	Get GL Report	Action Stage	Executes SQL query against NQRRECONGL - retrieves full GL dataset for the period. Saves to collection: NQR_Recon_GL_Report.

16	Get Duplicates	Action Stage	Executes SQL query against NQRRECONGL to detect duplicate entries. Saves to collection: Duplicates.
17	Generate Report	Page Reference	Calls Generate Report page - compiles all nine collections into a single Excel workbook with one worksheet per category.
18	Add Attachment Row	Action Stage	Adds the output workbook path to the email attachments collection.
19	Build Email Body	Multi-Calculation	Constructs the stakeholder email body including reconciliation summary and output file path.
20	Send Results Email	Action Stage	Sends the reconciliation workbook to all configured stakeholders.
21	Send Error Message	Action Stage	Dispatches an exception notification email if any unhandled error occurs during the run.
22	Block 1 - Data Items	Block Stage	Houses all data items and collections used by the process.
23	Block 2 - Exception Handler	Block Stage	Exception handler - sends appropriate alert email on any exception during the run.
24	Is Completed	Decision Stage	Final completion check - sets IsCompleted flag to True if all tasks completed successfully.
25	End	End Stage	Terminates the process - releases all resources cleanly.

5. Generate Report Page

5.1 Overview

The Generate Report page compiles all nine output collections into a single Excel workbook. Each collection is written to a dedicated worksheet in sequence, ensuring all reconciliation categories are clearly separated for ease of analysis and downstream processing.

5.2 Input Parameters

Input Parameter	Data Type	Description
OutputFileName	Text	File name of the Excel reconciliation workbook to be generated.
OutputFileFolder	Text	Folder path where the workbook will be saved.
Collection	Collection	The data collection to be written to the worksheet.

5.3 Generate Report Stages

S/N	Stage Name	Type	Description
1	Create Directory	Action Stage	Creates the output folder if it does not already exist.
2	Set Output File	Calculation Stage	Evaluates and sets the full output file name for the workbook.
3	Create Instance	Action Stage	Creates an Excel application instance.
4	Create Workbook	Action Stage	Creates a new Excel workbook.
5	Create Worksheet	Action Stage	Creates a worksheet for each output category.
6	Collection to Excel	Action Stage	Writes seven collections to dedicated worksheets: Result_Exception, Result_Matched, AutoReversed, GL_Exceptions, NQR_Settlement_Report, NQR_Recon_GL_Report, Duplicates.
7	Save Current Workbook	Action Stage	Saves the completed workbook to the OutputFilePath.
8	Close Workbook	Action Stage	Closes the Excel workbook and releases the instance.

6. Reconciliation Engine - spNQRRecon

6.1 Overview

The stored procedure spNQRRecon is the core of the NQR Reconciliation automation. It accepts a date range and performs all matching, classification, and exception handling logic entirely within the database engine - returning structured results to nine output tables that the Blue Prism bot then reads and compiles into the reconciliation workbook.

The procedure is executed by the bot via a SQL action stage - passing @start_date and @end_date in yyyy-MM-dd format. All matching decisions, exception classifications, and auto-reversal detections are encoded within the procedure, ensuring consistent and auditable reconciliation logic.

6.2 Processing Sequence

Step	Operation	Description
1	Create temp tables	Creates #tempMatchedItems, #ReversedMatchedItems, and #NQRRECONExceptions in memory for staging matched and reversed entries during processing.
2	Truncate result tables	Truncates NQRReconMatched and NQRRECONGLException to clear previous run results.
3	Load settlement data to temp	Reads NQRTransactionReport into #nqrtransactionreport, reformatting NIP_SESSION_ID (removing quotes) and PAYMENT_REFERENCE (removing NQR/ prefix) for matching.
4	Split GL by entry type	Reads NQRRECONGL into two temp tables: #NQRRECONGLCREDIT (PART_TRAN_TYPE = 'C') and #NQRRECONGLDEBIT (PART_TRAN_TYPE = 'D') - filtered to the transit account.
5	Detect auto-reversals	Identifies GL entries where a credit and debit share the same TRAN_RMKS - indicating automatic reversal. Both entries are inserted into #ReversedMatchedItems and removed from the credit/debit temp tables before matching begins.
6	Match settlement vs GL credit	Joins #nqrtransactionreport to #NQRRECONGLCREDIT on NIP_SESSION_ID = TRAN_RMKS or PAYMENT_REFERENCE = TRAN_RMKS. Matched pairs are inserted into #tempMatchedItems with Status = 'Matched'. Matched entries are then deleted from both temp tables.
7	Persist matched results	Inserts all records from #tempMatchedItems into the permanent NQRReconMatched table.
8	Classify GL exceptions	Any remaining entries in #NQRRECONGLCREDIT are inserted into NQRRECONGLException with status 'Cr Exception'. Any remaining in #NQRRECONGLDEBIT are inserted with status 'Dr Exception'.
9	Persist auto-reversals	Inserts all records from #ReversedMatchedItems into the permanent NQRRECONAutoReversed table.

10	Return settlement exceptions	Selects all remaining records from #nqrtransactionreport (entries that could not be matched to any GL entry) as the settlement exception output.

6.3 Matching Logic

The primary matching join uses two conditions - the bot matches a settlement entry to a GL credit entry if either the NIP_SESSION_ID equals the TRAN_RMKS, or the cleaned PAYMENT_REFERENCE equals the TRAN_RMKS. This dual-key matching handles the two reference formats used in the NQR payment scheme.

Auto-reversal detection uses the TRAN_RMKS field - if a GL credit and GL debit share the same remarks (exact match or contains relationship), both entries are classified as auto-reversed and excluded from the primary matching pass. This prevents reversed entries from appearing incorrectly as GL exceptions.

6.4 Key Database Tables

Table	Type	Purpose
NQRTransactionReport	Staging	Settlement report entries - truncated and reloaded at the start of each run from the shared folder files.
NQRRECONGL	Staging	GL transit account entries - truncated and reloaded from the GL query each run.
NQRReconMatched	Result	Successfully matched settlement-to-GL pairs - truncated and repopulated by spNQRRecon each run.
NQRRECONGLException	Result	Unmatched GL entries - credit exceptions and debit exceptions - truncated and repopulated each run.
NQRRECONAutoReversed	Result	Auto-reversed GL entry pairs detected during the matching pass.

7. Database Design

7.1 BNProducts Table

Master configuration table for the NQRR product. Read by the bot to determine product settings, schedule, and retry configuration.

Column	Data Type	Description
ProductCode	NVARCHAR	Unique product identifier - NQRR for NQR Reconciliation.
ProductName	NVARCHAR	Descriptive product name.
ActiveStatus	BIT	1 = Active. Setting to 0 disables the product without deletion.
StartTime	TIME	Scheduled start time - 08:10 AM for NQRR.
RunsOnNonBusinessDays	BIT	0 - NQRR runs on business days only.
MaxNoOfRetrials	INT	Maximum number of reconciliation attempts before the session is abandoned.
OutputFilePath	NVARCHAR	Network path where the bot writes its output reconciliation workbook.
IsPostingOnForReconciliation	BIT	Enables/disables automatic reconciliation posting.
From_Account_ID	NVARCHAR	Originating account for transactions.
STAN	NVARCHAR	System Trace Audit Number - unique transaction identifier.

7.2 BNProductList Table

Daily run tracker - populated at 02:00 AM. Updated by the bot throughout processing.

Column	Data Type	Description
ProductCode	NVARCHAR	Unique identifier - links to BNProducts.
HasProductCompleted	BIT	Flag indicating whether the reconciliation run completed successfully.
HasPostedCompleted	BIT	Flag indicating whether reversal posting completed successfully.
HasPostedReconciliation	BIT	Flag indicating whether reconciliation posting completed successfully.
ReconDate	DATE	Reconciliation date for this run instance.
NoOfRetries	INT	Number of processing attempts made by the bot for this session.

7.3 Dynamic File Path Structure

Token	Meaning	Example (6 Jan 2025)
dd	Day with leading zero	06
MMM	Abbreviated month	Jan
yyyy	Four-digit year	2025
[ ]	Transaction date bracket	[03 JAN 2025] = transaction date component
< >	Settlement date bracket	<06 JAN 2025> = settlement date component
( )	Transaction date + 1 bracket	(04 JAN 2025) = day after transaction date

8. Exception Handling

8.1 Overview

Exceptions are isolated to the current reconciliation run - a failure does not affect other products in the BN Operations Process loop. Every exception triggers an automated email alert to the operations and support teams.

8.2 Exception Types

S/N	Exception	Cause	Resolution
1	Settlement file not found	One or more required NQR report files absent from shared folder - may indicate upstream settlement session did not complete successfully.	Ensure all NQR settlement sessions for the day have completed. Verify files are in the shared folder and re-trigger.
2	Business Day API failure	API unavailable or returned an error.	Verify API service availability. Re-trigger after recovery.
3	GL not balanced	GL validation step found that spooled GL entries are not balanced or closing balance is unavailable.	Investigate GL data quality for the period. Verify GL extract is complete before re-triggering.
4	spNQRRecon execution failure	Stored procedure encountered a SQL error during reconciliation.	Review SQL error in Control Room logs. Check staging table data quality. Re-trigger after investigation.
5	Database connection error	BNRPA SQL Server unavailable.	Verify database availability. Re-trigger after recovery.
6	Wrong number of files	Fewer than expected files in shared folder - standard day expects 5 files, post-weekend expects 11.	Verify all NQR sessions ran. Obtain missing files and re-trigger.
7	Retry count exceeded	NQRR session has reached the maximum configured retry attempts.	Reset NoOfRetries in BNProductList and re-trigger after root cause investigation.

9. Operational Control and Alerting

9.1 Scheduling

Component	Schedule	Purpose
Load Products Job	02:00 AM daily	Pre-populates BNProductList for the day.
Load Products File List Job	02:00 AM daily	Pre-populates BNProductFileList with required settlement files.
Process Administrator	06:30 AM daily	Triggers BN Operations Process.
NQRR Reconciliation Run	08:10 AM daily	NQR Reconciliation session.

9.2 Alert Notifications

Alert Trigger	Type	Recipients
Settlement files not found in shared folder	Email	Operations Team, RPA Support
GL not balanced or validation failed	Email	Operations Team, RPA Support
Non-business day - reconciliation not required	Email	Operations Team
Reconciliation completed - results workbook	Email + Excel attachment	All configured stakeholders
Unhandled exception during run	Email	Operations Team, RPA Support

9.3 Failover Procedures

9.3.1 Manual Trigger

15. Log into Blue Prism Control Room.
16. Drag runtime resource to the BN Operations Process.
17. Click Start to initiate the session manually.

9.3.2 File Movement Alternative

18. Retrieve expected filenames from the exception email.
19. Navigate to the source file path specified in the email.
20. Move files to the path indicated by the bot.
21. Rename files to match the exact filenames expected by the bot.
22. Re-trigger via Control Room.

9.3.3 Configuration Changes

Change Required	Table	Action

Change reconciliation start time	BNProducts	UPDATE BNProducts SET StartTime = [new time] WHERE ProductCode = 'NQRR'
Disable NQRR temporarily	BNProducts	UPDATE BNProducts SET ActiveStatus = 0 WHERE ProductCode = 'NQRR'
Reset retry count	BNProductList	UPDATE BNProductList SET NoOfRetries = 0 WHERE ProductCode = 'NQRR' AND ReconDate = [date]

10. Security and Credentials

10.1 External Security

The Blue Prism environment and all connected systems are accessible only within the organisation's internal network. No bot components or reconciliation data are exposed to public networks.

10.2 Internal Security - Role-Based Access

S/N	Role
1	Release Managers
2	Developers
3	Process Administrators
4	Alert Subscribers
5	Web Service Consumers
6	Testers
7	Schedule Managers
8	Runtime Resources
9	Decipher User
10	System Administrators

10.3 Data Privacy and Preservation

- All reconciliation data held in bot memory is purged at the end of each run.
- Passwords stored in Blue Prism are encrypted and never written to logs.
- Stage logging configured to log errors only - no transaction data written to logs.
- Reconciliation results are stored in the BNRPA database for audit and traceability.
- Output workbooks archived in the designated output folder with date-stamped filenames.

11. Assumptions and Dependencies

11.1 Assumptions

- All NQR settlement report files are available in the shared folder before the 08:10 AM run time.
- Settlement records are complete and accurate.
- All NQR settlement sessions for the day have completed before the reconciliation run begins.
- The Business Day API returns accurate data for the current date.

11.2 External Dependencies

Dependency	Description
Business Day API	Called at the start of each run. If unavailable, the session cannot proceed.
Shared Network Folder	Source for all NQR settlement report files. Must be accessible and populated before the scheduled run time.
SMTP Email Service	Required for sending reconciliation reports and exception alerts to stakeholders.

11.3 Internal Dependencies

Dependency	Description
Runtime Resource	Bot machine must be online and Blue Prism runtime agent running for the session to execute.
BNRPA Database	Must be available for all staging, reconciliation, and result table operations.
spNQRRecon stored procedure	Must be deployed and accessible in the BNRPA database. Changes to the procedure require regression testing before production deployment.
Database Jobs	Load Products and Load Products File List jobs must complete at 02:00 AM before the 08:10 AM session.

12. Glossary

Term	Definition
NQR	National QR Code Payment - a payment scheme enabling QR-code-based merchant transactions.
GL	General Ledger - the bank's core accounting system recording all financial entries.
spNQRRecon	SQL stored procedure orchestrating the full NQR reconciliation - matching, auto-reversal detection, and exception classification.
Transit Account	The GL account through which NQR settlement entries are posted - source of GL data for reconciliation.
Open Item	A transaction that could not be reconciled - present in the GL but not in the settlement report, or vice versa.
Auto Reversed	A GL entry pair where a credit and debit share the same transaction remarks - indicating automatic reversal.
NQRR	Product code for the NQR Reconciliation process.
Runtime Resource	A machine with Blue Prism installed from which the bot executes. Must be online for sessions to run.
Settlement Date	The day on which the processor provides settlement for transactions initiated.
Transaction Date	The day on which the underlying transaction was performed.
DB	Database - specifically the BNRPA SQL Server database used by this automation.

- End of Document -